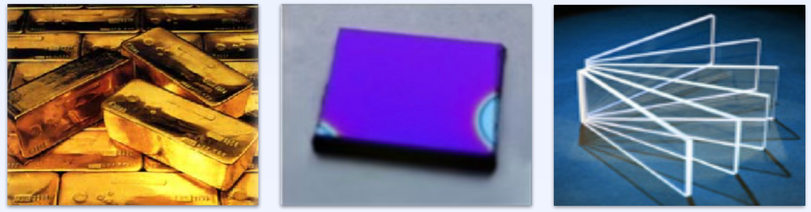
Tailoring Near-Infrared Epsilon-Near-Zero Optical Resonances in Conducting Polymer Thin Films via Controlled Processing

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Introduction & motivation

Epsilon-near-zero (ENZ) optical materials

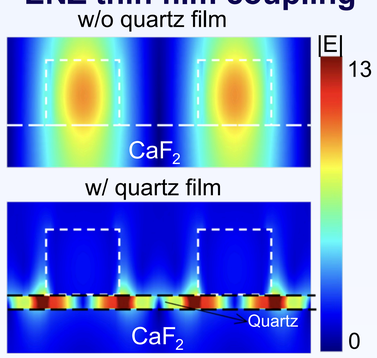


Re[ε] < 0 | Re[ε] ≈ 0 | Re[ε] > 0

- At the ENZ wavelength, Re[ε] ≈ 0. [1] The electric field can be confined and enhanced within a thin film. [2]

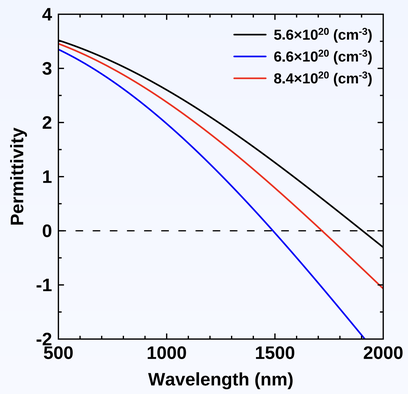
$$\varepsilon_i E_{i,\perp} = \varepsilon_{\mathrm{ENZ}} E_{\mathrm{ENZ},\perp}$$
$$E_{\mathrm{ENZ},\perp} = \frac{\varepsilon_i}{\varepsilon_{\mathrm{ENZ}}} E_{i,\perp}$$

ENZ thin film coupling



Yue et al. (2024) [3]

ENZ v.s. carrier density

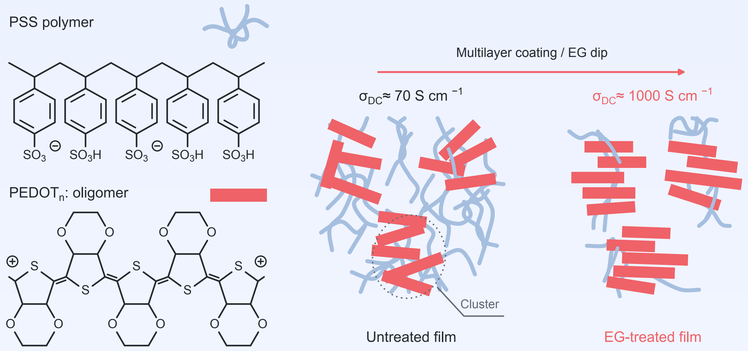


Motivation:

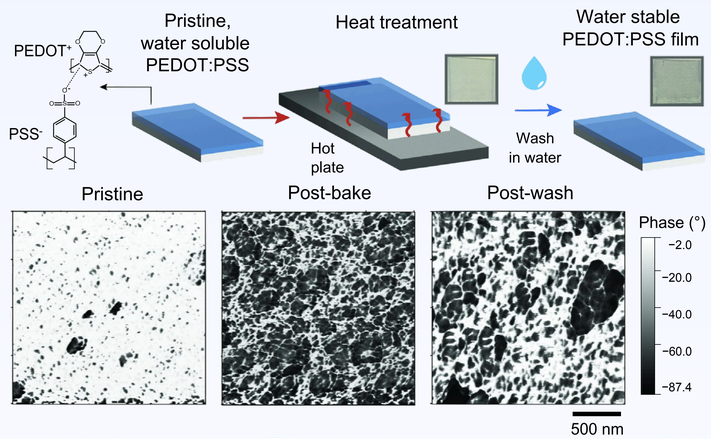
- Precise control on ENZ wavelength and thickness.
- Electrical tunability of ENZ after fabrication.
- Achieve ENZ materials for visible wavelength with low loss

Towards controllable ENZ

Ethylene glycol (EG) assisted processing



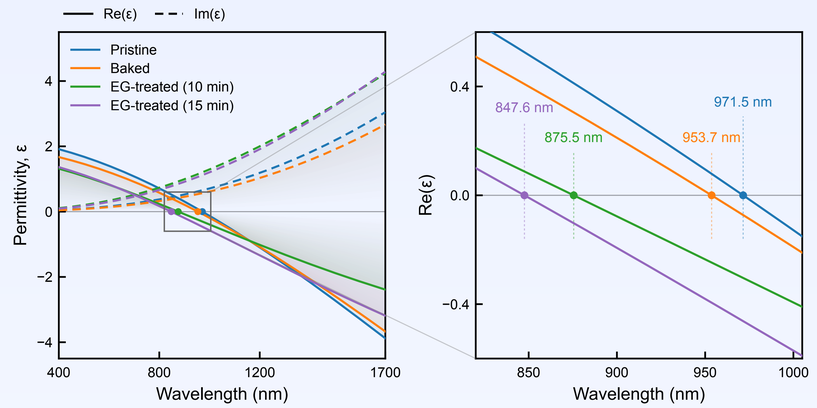
Thermal processing



- **Chemical Processing:** EG-assisted processing reduces PSS-rich barriers and improves transport. conductivity: ~70 → ~950 S cm^{-1}. [6]
- **Thermal Processing:** Heat reorganizes PEDOT/PSS domains; water washing removes excess PSS. AFM: 180 °C, 2 min.

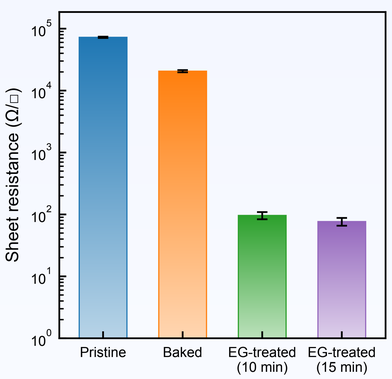
Experimental results

Ellipsometry-derived permittivity and ENZ blue shift

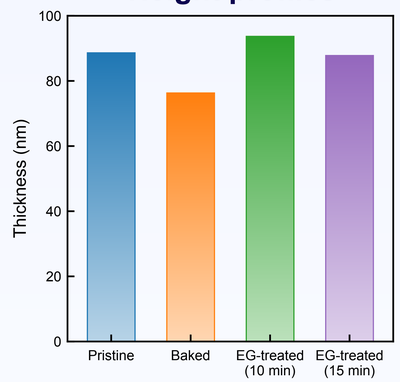


- **ENZ wavelength:** Pristine **971.5 nm**; Baked **953.7 nm**; EG-treated (10 min) **875.5 nm**; EG-treated (15 min) **847.6 nm**

Sheet resistance



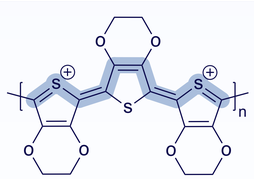
Height profiles



- Sheet resistance decreases after baking, with a much larger reduction observed after EG treatment.

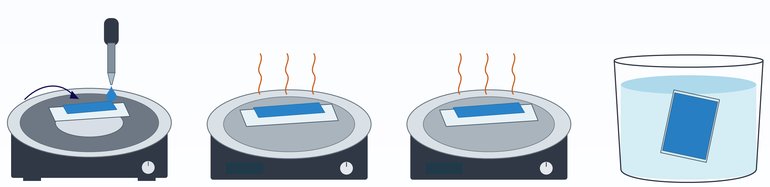
Conjugated polymer — PEDOT:PSS

- **PEDOT:PSS** is a solution-processable and conductive polymer [4] whose optical and electrical **properties** can be dynamically **controlled** through its **oxidation state.** [5]
- Its combination of high conductivity, optical transparency and reversible redox activity has enabled various applications in active photonics, bioelectronics, and electrically switchable optical devices. [4],[6],[7]



Fabrication & ENZ evaluation

Spin coating	Drying	Baking	EG Processing
3000 rpm for 30s	80°C for 30 mins	200 °C for 1 h	EG immersion 10mins



Outlook & Impact

- **Laser writing:** Explore femtosecond laser writing to pattern PEDOT:PSS microstructures and investigate spatial control of its ENZ response
- **Nonlinear photonics:** ENZ field enhancement boosts third-harmonic generation, enabling frequency conversion in ultrathin films.
- **Tunable organic metasurface:** Use laser processing to fabricate PEDOT nanopillar arrays that form an ENZ metasurface.

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